

- In each case find the values of expression in Matlab and explain precisely the order in which calculation was performed:-
 - $-2^3 + 9$
 - $\frac{2}{3} * 3$
 - $3 * (\frac{2}{3})$
 - $3 * 4 - 5^2 * 2 - 3$
 - $(\frac{2}{3^2}) * (3 - 4^3)^2$
 - $3 * (3 * 4 - 2 * 5^2 - 3)$
- Write in Matlab code:-
 - $\frac{(2+6i)}{(9i)(1+i)}$
 - Find the absolute value and angle in Matlab of part (a).
- Given array $A = [2, 4, 1; 6, 7, 2; 3, 5, 9]$ provide the commands needed to:-
 - Assign the first row of A to vector called x .
 - Assign the last two row of A an array called y .
 - Compute the sum over the column of A .
- Let $X = [3, 2, 6, 8]^T$ and $Y = [4, 1, 3, 5]^T$ add X and Y then find sum of elements of resulting vector.
- Let $A = [1, 2, 3, 4; 5, 6, 7, 8; 1, 1, 2, 3]$ then write in Matlab:-
 - To extract a sub-matrix B consisting of row 1 and 3, column 2 and 4.
 - Inter change row 1 and 3.
 - Find $A(:)$. Explain what it does?
 - Delete 2^{nd} column of A and delete 3^{rd} row of A .
 - Remove the second column.
- Suppose vectors a and b are defined as follows:

$$a = [2, -1, 5, 0];$$

$$b = [3, 2, -1, 4];$$
 Evaluate by hand the vector c in the following statements. Check your answers in Matlab.
 - $c = a - b$.
 - $c = +a - 3b$.
 - $c = 2 * a + a.^b$.
 - $c = b./a$.
 - $c = b.a$.
 - $c = 2 * b/3. * a$.
 - $c = b * 2. * a$.
- Given a script for animating the Mexican hat problem.


```
[x,y] = meshgrid (-8 : 0.5 : 8);
r = sqrt( x.^2 + y.^2 ) + eps;
z = sin(r)./r;
mesh(z);
```

 type this into the editor, save it, and execute it.
- Try to avoid using unnecessary brackets in an expressions. Can you spot the errors in the following expression?

$$(2(3+4)/(5*(6+1))^2$$
- Use Matlab array operations to do the following :
 - Add 1 to each element of the vector $[2, 3, -1]$.
 - Multiply each element of the vector $[1, 4, 8]$ by 3.
 - Find the array product of the two vector $[1, 2, 3]$ and $[0, -1, 1]$.
 - square each element of the vector $[2, 3, 1]$.

1. Create 3-by-3 matrix m , then copy the second and third rows of the matrix twice to create 4-by-3 matrix, then
 - (a) Delete 2^{nd} row.
 - (b) Find sub-matrix of order 2-by-2.
2. Create two matrix of order 3-by-3, add, subtract both matrices-
 - (a) Find multiplication of the both matrix.
 - (b) Find element-wise multiplication of both matrix and see the difference from above multiplication matrix.
3. Given $A = [2, 4, 1; 6, 7, 2; 3, 5, 9]$ and $B = [7, 5, 6; 2, 0, 8; 5, 7, 1]$, find the left division and right division of the matrix,
4. Let $A = [10, 12, 23; 14, 8, 6; 27, 8, 9]$ then perform the scalar operations on the matrix A by scalar quantity $b = 5$.
5. Let $A = [10, 12, 23; 14, 8, 6; 27, 8, 9]$ and $B = [12, 31, 45; 8, 0, -9; 45, 2, 11]$, then find the horizontal concatenation and vertical concatenation of the given matrices.
6. Find the following matrices of the order 3-by-3 :
 - (a) Identity matrix.
 - (b) Zero Matrix.
 - (c) Random number matrix and magic matrix.
7. Create a two dimensional array matrix say A then add a third dimension to A .
8. Perform the relational operators to the given single value -
 - (a) If $a=3$ and $b = 4$
 - (b) If $a= 10$ and $b=5$.
9. Perform the logical operators (AND,OR,NOT) to the given single value -
 - (a) If $a= 10$ and $b = 0$
 - (b) If $a= 2$ and $b = 5$.
10. Perform the logical operator and relational operator to find the relation is 'TRUE' or 'FALSE'-
 - (a) $2+3==5$ OR $3 > 7$.
 - (b) $4+5 == 3$ AND $4 == 4$.
 - (c) NOT $7 > 3$.
11. Define a vector of length 100 between the point 1 and 2, also find the computational time.

- Use the dot function to find the dot product of the following vectors:
 - $\vec{A} = [31, 2, 3, 44]$ and $\vec{B} = [12, 20, 15, 74]$
 - Find the dot product of $\vec{A}$ and $\vec{B}$ by summing the array products of $\vec{A}$ and $\vec{B}$.
- Consider the following matrix $A = [1, 2, 3; 2, 4, 6; 3, 6, 9]$. Would you expect it to be singular or not? If it non-singular find the inverse of matrix A .
- Consider the following system of three equations with three unknowns:

$$\begin{aligned}
 (a) \quad & \begin{aligned} x + y + z + w &= 10 \\ 2x + 3y + z + 5w &= 31 \\ -x + y - 5z + 3w &= -2 \\ 3x + y + 7z - 2w &= 18. \end{aligned} \\
 (b) \quad & \begin{aligned} 3x + 2y - z &= 10 \\ 2x + 3y + z &= 31 \\ -x + 3y + 2z &= -2 \\ x - y - 7z &= 18. \end{aligned}
 \end{aligned}$$

Find the solution of the system.

- Solve the following systems using left division:
 - $$\begin{aligned} 3x + 2y + 5z &= 22 \\ 2x + 3y - 2z &= 8 \\ x + y + z &= 6 \\ 2x - 4y - 7z &= -27. \end{aligned}$$
 - $$\begin{aligned} 3x + 2y + 5z &= 22 \\ 4x + 5y - 2z &= 8. \end{aligned}$$
- Compute the matrix product $A*B$ of the following pairs of matrices:
 - $A = [12, 4; 3, -5]$ and $B = [2, 12; 0, 0]$.
 - $A = [1, 2, 3; 2, 4, 6]$ and $B = [-2, 4; 3, 8; 12, 2]$.

Show that $A * B$ is not the same as $B * A$.

- Solve the following systems of equations, using both matrix left division and the inverse matrix method:
 - $-2x + y = 3, x + y = 10.$
 - $$\begin{aligned} 5x + 3y - z &= 10 \\ 3x + 2y + z &= 4 \\ 4x - y + 3z &= 12. \end{aligned}$$
 - $$\begin{aligned} 3x + 2y + 5z + w &= 24 \\ x - 3y + 7z + w &= 12 \\ 2x + 2y - 3z + 4w &= 17 \\ x + y + z + w &= 0. \end{aligned}$$
- In general, matrix left division is faster and more accurate than the matrix inverse. Using both techniques, solve the above system of equations and time the execution with the tic and toc functions.
- The future-value-of-money formula relates how much a current investment will be worth in the future, assuming a constant interest rate.

$$FV = PV * (1 + I)^n,$$

where FV is the future value, PV is the present value or investment, I is the interest rate expressed as a fractional amount per, compounding period-i.e., 5% is expressed as .05, n is the number of compounding periods.

- Create a MATLAB function called future value with three inputs: the investment (present value), the interest rate expressed as a fraction, and the number of compounding periods.
- Use your function to determine the value of a \$1000 investment in 10 years, assuming the interest rate is 0.5% per month, and the interest is compounded monthly.

9. This problem requires you to generate temperature-conversion tables. Use the following equations, which describe the relationships between temperatures in degrees Fahrenheit (T_F), degrees Celsius T_C , kelvins (T_K), and degrees Rankine (T_R), respectively:

$$T_F = T_R - 459.67^\circ R$$

$$T_F = 9/5T_C + 32^\circ F,$$

$$T_R = 9/5T_K$$

Create the primary function should be named as `temperature_conversions` and should include the subfunctions `F_to_K`, `C_to_R`, `C_to_F`.

Within the primary function use the subfunctions to:

- Generate a conversion table for values from $0^\circ F$ to $200^\circ F$. Include a column for temperature in Fahrenheit and Kelvin.
- Generate a conversion table from $0^\circ C$ to $100^\circ C$. Print 25 lines in the table. (Use the `linspace` function to create your input vector.) Your table should include a column for temperature in Celsius and Rankine.
- Generate a conversion table from $0^\circ C$ to $0^\circ C$. Choose an appropriate spacing. Include a column for temperature in Celsius and Fahrenheit.

10. A rocket is launched vertically. At time $t = 0$, the rocket's engine shuts down. At that time, the rocket has reached an altitude of 500 m and is rising at a velocity of 125 m/s. Gravity then takes over. The height of the rocket as a function of time is

$$f(t) = -4.9t^2 + 100t + 250 \quad t > 0$$

- Create a function called `height` that accepts time as an input and returns the height of the rocket. Use your function in your solutions to parts b and c.
 - Find the time when the rocket starts to fall back to the ground. (The `max` function will be helpful in this exercise.)
11. (a) Create an anonymous function called `my_function`, equal to

$$-x^5 - 5x^3 - 3x^2 + 3 + e^x$$

- Use the `fminbnd` function to find the minimum function value in this range. The `fminbnd` function is an example of a function function, since it requires a function or function handle as input. The syntax is `fminbnd(function_handle, xmin, xmax)`.

Three inputs are required: the function handle, the minimum value of x , and the maximum value of x .

The function searches between the minimum value of x and the maximum value of x for the point where the function value is a minimum.

1. Plot x versus y for $y = \sin(x)$. Let x vary from 0 to 2π in increments of 0.1π .
 - (a) Add a title and labels to your plot.
2. Plot x versus y_1 and y_2 for $y_1 = \sin(x)$ and $y_2 = \cos(x)$. Let x vary from 0 to 2π in increments of 0.1π . Add a title and labels to your plot.
 - (a) Add a legend to the graph.
 - (b) Adjust the axes so that the x -axis goes from -1 to $2\pi + 1$ and the y -axis from -1.5 to $+1.5$.
3. Subdivide a figure window into two rows and one column.
 - (a) plot $y = \tan(x)$ for $-1.5 \leq x \leq 1.5$. Use an increment of 0.1.
 - (b) Add a title and axis labels to your graph.
 - (c) plot $y = \sinh(x)$ for the same range.
 - (d) Add a title and labels to your graph.
4. Create a plot of the functions that follow, using **fplot**.
 - (a) $f(t) = 5t^2$.
 - (b) $f(t) = 5(\sin(t))^2 + t(\cos(t))^2$.
 - (c) $f(t) = \sin(t) + \log(t)$.
5. Create a vector x of values from 0 to 20π , with a spacing of $\frac{\pi}{100}$. Define vectors y and z as

$$y = x \sin(x)$$

and

$$z = x \cos(x)$$

- (a) Create an x-y plot of x and y .
 - (b) Create a polar plot of x and y .
 - (c) Create a three-dimensional line plot of x , y , and z . Don't forget a title and labels.
 - (d) Generate a contour plot of Z .
 - (e) Generate a combination surface and contour plot of Z .
6. Create x and y vectors from -5 to $+5$ with a spacing of 0.5. Use the meshgrid function to map x and y onto two new two-dimensional matrices called X and Y . Use your new matrices to calculate vector Z , with magnitude

$$Z = \sin(\sqrt{X^2 + Y^2})$$

- (a) Use the mesh plotting function to create a three-dimensional plot of Z .
 - (b) Use the surf plotting function to create a three-dimensional plot of Z . Compare the results you obtain with a single input Z with those obtained with inputs for all three dimensions (X, Y, Z).
7. Let the vector $G = [68, 83, 61, 70, 75, 82, 57, 5, 76, 85, 62, 71, 96, 78, 76, 68, 72, 75, 83, 93]$ represent the distribution of final grades in an engineering course.
 - (a) Use MATLAB to sort the data and create a bar graph of the scores.
 - (b) Create a histogram of the scores, using hist.
 8. Draw the mesh plot of following function:
 - (a) $z = \frac{xe^x}{2y}$ for $1 \leq x \leq 5$ and $1 \leq y \leq 5$.
 - (b) $z = \log(x^2 + y^2)$ for $1 \leq x \leq 5$ and $1 \leq y \leq 5$.
 - (c) $z = \frac{y^P 2}{x^2 + y^2}$ for $1 \leq x \leq 5$ and $1 \leq y \leq 5$

1. Given two vectors, x and y . The x values begin at 0 and end at 15π , with a spacing of $\frac{\pi}{100}$, while the y values begin at 0 and end at 12, with a spacing of $\frac{\pi}{10}$.
 - (a) Find size, bytes and class of x and y .
 - (b) Save input and output using 'diary'.
2. Create a vector x of values from 0 to 10π , with a spacing of $\frac{\pi}{100}$. Define vector y as

$$y = \pi \sin(x)$$

- (a) Find size, bytes and class of x and y .
 - (b) Save input and output using 'diary'.
3. Given $y1 = 2 \cos(x)$ and $y2 = 5 \sin(x)$. Let x vary from 0 to 2π in increments of 0.1π .
 - (a) Save and load workspace variables.
 - (b) Create new script, load the data saved in the workspace and plot x versus $y1$.
 - (c) Create new script, load the data saved in the workspace and plot x versus $y2$.(you may use 'doc' command for help)
 4. Given $y1 = 2x$ and $y2 = 5 \sin(x)$. Let x vary from 0 to π in increments of 0.1π .
 - (a) Save and load workspace variables.
 - (b) Create new script, load the data saved in the workspace and plot x versus $y1$.
 - (c) Create new script, load the data saved in the workspace and plot x versus $y2$.
 5. Create a vector x of values from 0 to 2π , with a spacing of $\frac{\pi}{4}$. Define vector y as

$$y = \sin(x)$$

Next, define a new vector, xq of values from 0 to 2π , with a spacing of $\frac{\pi}{16}$.

- (a) Find the yq values corresponding to the xq values by linear interpolation.
 - (b) On the same figure, plot the original y vs. x as circles, and yq vs. xq as a line.
 - (c) Repeat the exercise in part (a) and (b) using the `spline(...)` function to interpolate.
6. Create a vector x of values from 1 to 9. Define vector v as

$$v = [0 \ 1.41 \ 2 \ 1.41 \ 0 \ -1.41 \ -2 \ -1.41 \ 0]$$

.

Next, define a new vector xq of values from 1.5 to 8.5.

- (a) Find the yq values corresponding to the xq values by linear interpolation.
- (b) On the same figure, plot the original y vs. x as circles, and yq vs. xq as a line.
- (c) Repeat the exercise in part (a) and (b) using the `spline(...)` function to interpolate.

1. A factorial is the product of all the integers from 1 to N . In Matlab, find the Factorial for $N = 5$.

2. Use a *for* loop to create a vector containing the first 10 elements in the harmonic series, *i.e.*,

$$\frac{1}{1} \quad \frac{1}{2} \quad \frac{1}{3} \quad \cdots \quad \frac{1}{10}$$

3. Use a *for* loop to create a vector containing the first 10 elements in the alternating harmonic series, *i.e.*,

$$\frac{1}{1} \quad -\frac{1}{2} \quad \frac{1}{3} \quad -\frac{1}{4} \cdots -\frac{1}{10}$$

4. Write Matlab code to calculate the sum of series:

$$S = 1 - \frac{x}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} \quad \text{for } x = 1.5.$$

5. Write the Matlab program to obtain the sum of all even number from 0 to 20 using *for* loop statement.

6. Write the program to find the average value of given any 10 number using *for* loop statement. Check the program with the following values : 35, 24, 5, 6, 4, 10, 23, 45, 2

7. Repeat the preceding problem, this time using a *while* loop.

8. Write a program, using *while* loop, for finding square of integers less than 5.

9. Write the Matlab program, to test whether or not π^e is greater than, or equal to, e^π .

10. In Matlab, compute the Fibonacci sequence defined by $f_1 = 0, f_2 = 1$ and $f_n = f_{n-1} + f_{n-2}$. Using *for* loop and *if* condition.

11. Draw graphs of $\sin(nx)$ on the interval $[-1, 1]$ for $n = 1, 2, 3, 4, 5, 6, 7, 8$.

12. For any n build the $n \times n$ matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \cdots & 0 \\ 1 & 1 & 0 \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 1 \end{bmatrix}$$

in Matlab.

1. Use the *syms* or *syms* command to create the following symbolic expressions:

- (a) $ex1 = x^2 - 1$
- (b) $ex2 = a * x^3 + b * x^2 + c * x + d$
- (c) $ex3 = \sin(x)$
- (d) $eq1 = x^2 = 1$
- (e) $eq2 = a * x^2 + b * x + c = 0$
- (f) $eq3 = (X + 1)^2 = 0$
- (g) $a = 6/5 + 4/3$, using both numerical and symbolic computation.

2. Use the variables defined in Practice Exercises 1 in these exercises.

- (a) Multiply $ex1$ by $ex2$, and name the result $y1$.
- (b) Divide $ex1$ by $ex2$, and name the result $y2$.
- (c) Use the *numden* function to extract the numerator and denominator from $y1$ and $y2$.
- (d) Use the *factor*, *expand*, *collect*, and *simplify* functions on $y1$ and $y2$.

3. Define a symbolic equation representing each equation in the given system of equations. Use the *solve* function to solve for x , y , and z of the following equation:

$$\begin{aligned}5 * x + 6 * y - 3 * z &= 10 \\3 * x - 3 * y + 2 * z &= 14 \\28x - 4 * y - 12 * z &= 24\end{aligned}$$

4. Use the variables defined in Practice Exercises 1 in these exercises. Use the *solve* command to solve the $eq1$, $eq2$ and $eq3$.

5. Using the *subs* function, substitute 4 into each expression/equation defined in Practice Exercises 1 for x .

6. Create a symbolic function called g for the following:

$$x^2 + \sin(x) * x$$

and use it to evaluate $g('a')$, $g(3)$, and $g([1 : 5])$.

7. Plot the functions:

- (a) Use *ezplot* to plot $\cos(x)$ from -2π to $+2\pi$.
- (b) Use *ezplot* to create an implicit plot of $x^2 - y^4 = 5$.
- (c) Use *ezplot* to create a parametric plot of $x = \sin(t)$ and $y = 3 \cos(t)$.

8. Find the first derivative with respect to x of the following expressions:

- (a) $x^2 + x + 1$
- (b) $\sin(x)$
- (c) $\ln(x)$
- (d) $\tan(x)$

9. Find the first partial derivative with respect to x of the following expressions:

- (a) $ax^2 + bx + c$
- (b) $\sin(xy)$
- (c) $x^{\frac{1}{2}} - 3y$
- (d) $\tan(x + y)$

10. Integrate the following expressions with respect to x :

- (a) $3x^3 - 5x^2 + 3$

(b) $4x^3 + \cos(2x) + 3$

(c) $4e^{-xt} + \sin(3t)$

(d) $\sin(3x) + x \cos(x)$

11. Integrate the following expressions with respect to x :

(a) $3x^3 - 5x^2 + 3$; limits are $(3, 5)$

(b) $9xe^x$; limits are $(0, 5)$

(c) $4e^{-xt} + \sin(3t)$; limits are $(0, \pi)$

(d) $\sin(3t) + x \cos(x)$; limits are $(0, \pi)$

1. Calculate and plot the value of y using solver *ode45* at all point for the interval $t = 0$ to $t = 2$ if

- (a) $\frac{dy}{dt} = 12t^2 + 8t + 4$ with I.C. $y(0) = 2$
- (b) $\frac{dy}{dt} = y(e^t - 1)$ with I.C. $y(0) = 1$
- (c) $\frac{dy_1}{dt} = y_2 - 1$ and $\frac{dy_2}{dt} = y_1 - t$ with I.C. are $y_1(0) = 1$, $y_2 = 1$
- (d) $\frac{dy}{dt} = t - 0.1y^2$ with I.C. $y(0) = 1$, find the value of $y(0.8)$
- (e) $\frac{dy_1}{dt} = \frac{-1}{2}y_2 + \frac{y_1 y_2}{20}$ and $\frac{dy_2}{dt} = y_1 + \frac{y_1 y_2}{10}$ with I.C. are $y_1(0) = 1$, $y_2 = 1$.

2. Use the ode function defined in Practice Exercises 1 calculate and plot the value of y using *ode23* ode solver.

3. Calculate and plot the value of y using solver *ode45* and *ode23*, if $A = \begin{bmatrix} -5 & 0 & -100 \\ 2 & -2 & 0 \\ 0 & 0.1 & -0.08 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 0 \\ -0.1 \end{bmatrix}$, obtain the solution of the following set of differential equation for $t = [0, 8]$

$$\frac{dy}{dt} = Ay + Bu \text{ with I.C., } y(0) = [0, 0, 0]'$$

where $u = 0.2$.

4. Solve the following differential equation exactly using *dsolve* command in Matlab:

- (a) $\frac{dy}{dt} = ay$
- (b) $\frac{dy}{dt} = a^2 y$ with $y(0) = 1$
- (c) $\frac{dy}{dt} = \sin(t)$
- (d) $\frac{dy}{dt} = x$ and $\frac{dx}{dt} = y$
- (e) $\frac{dy}{dt} = x - 1$ and $\frac{dx}{dt} = y - t$ with $x(0) = 1$ and $y(0) = 1$
- (f) $\frac{dy}{dt} = y(e^t - 1)$ with $y(0) = 1$.